

Equality of seminorm balls under a linear bijection preserves the seminorms

MathlibAnnex.SeminormBall.map_eq

theorem

Turns a ball-image equality into one seminorm inequality in each direction.

Statement

Let E, F be real vector spaces with definite seminorms p and q , and write $B_p = \{x : p(x) \leq 1\}$ and $B_q = \{y : q(y) \leq 1\}$. If a real-linear bijection $L : E \rightarrow F$ satisfies $L(B_p) = B_q$, then

$$q(Lx) = p(x) \quad (x \in E).$$

Assumptions

The spaces are real vector spaces. The seminorms satisfy $p(x) = 0 \Rightarrow x = 0$ and $q(y) = 0 \Rightarrow y = 0$. The map L is linear and bijective, and its image of the entire closed seminorm unit ball is exactly B_q . No topology, continuity, measure, compactness, finite-dimensionality or volume equality is assumed.

Conclusion

The given map L preserves the two seminorms pointwise. With ordinary norms this gives the separately stated norm-preservation result for a linear bijection carrying one closed unit ball onto the other.

Notes

The auxiliary forward inequality only needs definiteness of the domain seminorm and containment of ball images. In the equality theorem, definiteness of q is used when that same inequality is applied to L^{-1} . The argument is algebraic.

Proof route

For $x = 0$, linearity and the seminorm axioms give equality. Suppose $x \neq 0$. Definiteness and nonnegativity give $p(x) > 0$. The normalized vector $u = x/p(x)$ satisfies $p(u) = 1$, so $u \in B_p$. The forward ball containment gives $L(u) \in B_q$. By linearity and seminorm homogeneity,

$$\frac{q(Lx)}{p(x)} = q(L(x/p(x))) \leq 1,$$

and multiplying by $p(x) > 0$ gives $q(Lx) \leq p(x)$. This is the ball-containment lemma, valid for every x after including zero. Since L is bijective and $L(B_p) = B_q$, its inverse carries B_q into B_p . Apply the same lemma with (q, p, L^{-1}) to Lx :

$$p(x) = p(L^{-1}(Lx)) \leq q(Lx).$$

Together the two inequalities give the desired equality.

▼ Proof steps

1. To verify the inverse containment, let $y \in B_q$. The image equality supplies $u \in B_p$ with $Lu = y$. Injectivity gives $L^{-1}y = u \in B_p$. Thus the reversed ball-containment lemma has all its hypotheses, including definiteness of q .
2. The two applications of the same lemma have opposite directions: $q(Lx) \leq p(x)$ and $p(L^{-1}(Lx)) \leq q(Lx)$. Substitute the inverse identity in the second and apply antisymmetry. No conclusion is drawn from measure equality alone.

Main citations

- Exact declaration and its source — `MathlibAnnex.SeminormBall.map_eq`
- The seminorm inequality from ball containment — `MathlibAnnex.SeminormBall.map_le`
- The norm-ball containment specialization — `MathlibAnnex.NormBall.map_le`
- The norm-ball equality specialization — `MathlibAnnex.NormBall.map_eq`

Lean source signature (exact)

```
theorem map_eq
  [AddCommGroup E] [Module ℝ E]
  [AddCommGroup F] [Module ℝ F]
  (p : Seminorm ℝ E) (q : Seminorm ℝ F)
  (hp : ∀ x : E, p x = 0 → x = 0)
  (hq : ∀ y : F, q y = 0 → y = 0)
  (L : E →ℝ F) (hbij : Function.Bijective L)
  (hball : L '' p.closedBall 0 1 = q.closedBall 0 1) (x : E) :
  q (L x) = p x
```

In the source	Mathematical meaning
<code>[AddCommGroup E] [Module ℝ E] [AddCommGroup F] [Module ℝ F]</code>	The algebraic real vector spaces E, F ; these binders specify no topology.
<code>(p : Seminorm ℝ E) (q : Seminorm ℝ F)</code>	The domain seminorm p and codomain seminorm q .
<code>(hp : ∀ x : E, p x = 0 → x = 0)</code>	Definiteness of p : $p(x) = 0$ implies $x = 0$ for every $x \in E$.
<code>(hq : ∀ y : F, q y = 0 → y = 0)</code>	Definiteness of q : $q(y) = 0$ implies $y = 0$ for every $y \in F$.

In the source	Mathematical meaning
<code>(L : E →_ℝ F) (hbij : Function.Bijective L)</code>	The same real-linear map $L : E \rightarrow F$ is injective and surjective. This map type does not bundle continuity.

In the source	Mathematical meaning
<code>(hball : L '' p.closedBall 0 1 = q.closedBall 0 1)</code>	The set-image equality $L(B_p) = B_q$, where $B_p = \{x : p(x) \leq 1\}$ and $B_q = \{y : q(y) \leq 1\}$. It supplies both directions of the ball comparison.
<code>(x : E) : q (L x) = p x</code>	At every supplied $x \in E$, the conclusion is $q(Lx) = p(x)$.

▼ Exact surrounding binder context (separate excerpt)

```

noncomputable section

open Set MeasureTheory
open scoped ENNReal

namespace MathlibAnnex

namespace SeminormBall

variable {E F : Type*}

```

▼ Exact content identity

Declaration: `MathlibAnnex.SeminormBall.map_eq`

Accepted content SHA-256: 0250218f816ad2452f53b342a8f021d1cc81d427d4768cb11e2beaf81f10df8f

Accepted source guide SHA-256: 7dbef9f97a6d1bdbecababbef6d8cdebb359e395f730d4a19671d8eb296f958a

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73